

交通大数据

主成分分析与因子分析

顾子渊， 副研究员、博导
复杂交通网络研究中心
ziyuangu@seu.edu.cn



主成分分析与因子分析

□ 学习目标

- 主成分分析
- 因子分析



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主成分分析

□ 隐变量模型 (latent variable model)

➤ 隐变量：不可观测或不易表达的变量

- ✓ 智商观察不到，但成绩可以观察到
- ✓ 成绩一定程度上反映了智商，或智商可由成绩来部分解释



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➤ 隐变量模型：

- ✓ **主成分分析（降维）**：将多个变量转化为少数几个综合变量（原变量的线性组合，即主成分），对原数据进行有效解释
- ✓ **因子分析（公约数）**：找出多个变量的共同影响因素，浓缩成少数几个因子变量



主成分分析

□ 原理

- 以**最少信息丢失**为前提，将原变量化为少数几个**互不相关**的综合变量（原变量的**线性组合**，即主成分）
 - ✓ 主成分个数远远少于原变量数
 - ✓ 主成分包含原变量绝大部分信息
 - ✓ 主成分互不相关
 - 若两个变量有一定相关关系 → 两个变量间信息有重叠
 - 物理、化学成绩 → 理科水平



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□ 目的

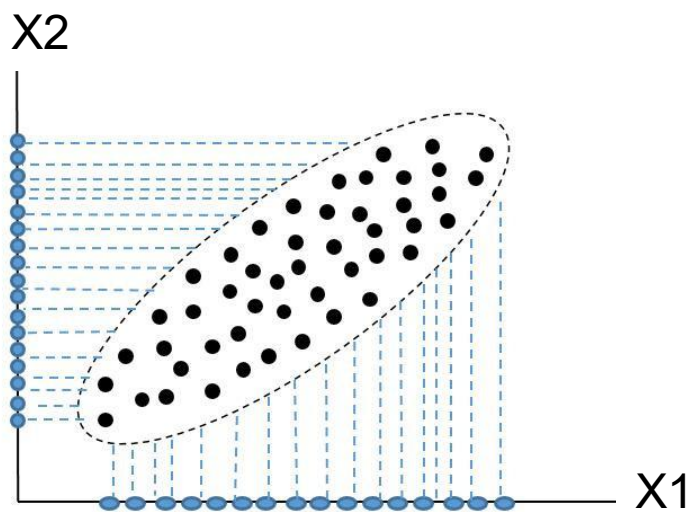
- **削减变量数，同时保留大部分信息** → 高效解释数据变化
 - ✓ 少数几个变量解释了90%的原数据变化 → 集约性 & 有效性



主成分分析

□ 数学推导

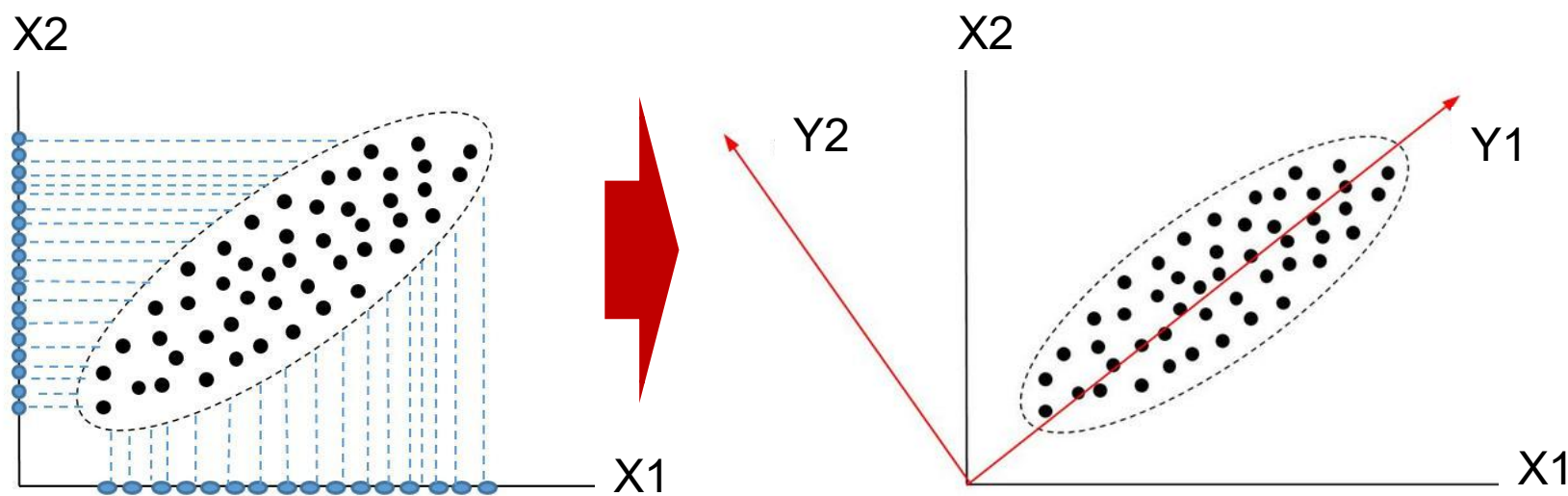
坐标旋转 → 降维



主成分分析

□ 数学推导

坐标旋转 → 降维

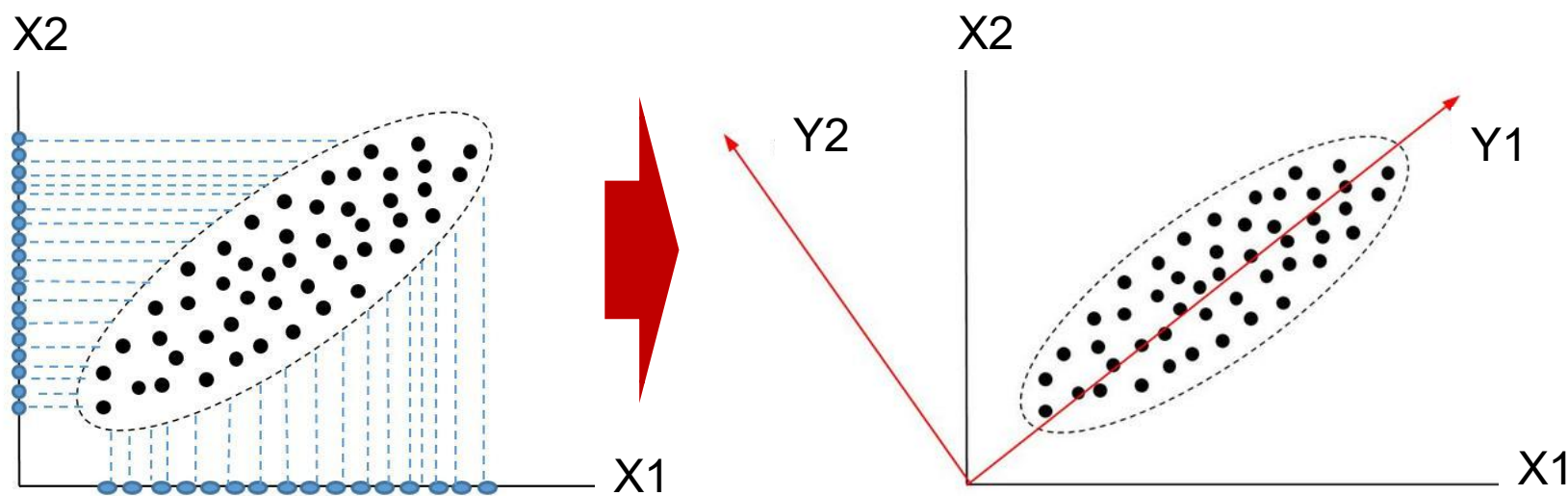


- Y1坐标轴投影重叠少、范围大：信息保留多
- Y1、Y2坐标轴投影正交：不相关

主成分分析

□ 数学推导

坐标旋转 → 降维



- Y1坐标轴投影重叠少、范围大：信息保留多
- Y1、Y2坐标轴投影正交：不相关
- **旋转多少度？Y1、Y2 是 X1、X2 线性组合？**

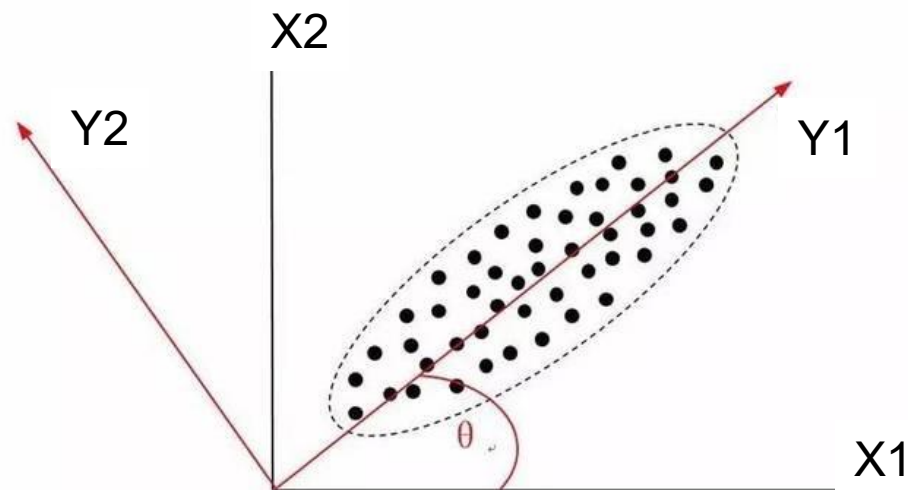
主成分分析

□ 数学推导

➤ 假设原坐标系旋转了 θ

$$\begin{aligned} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} &= \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{aligned}$$

$$y_i = \mathbf{u}_i^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad i = 1, 2$$



✓ 如何确定 u_{ij} ?

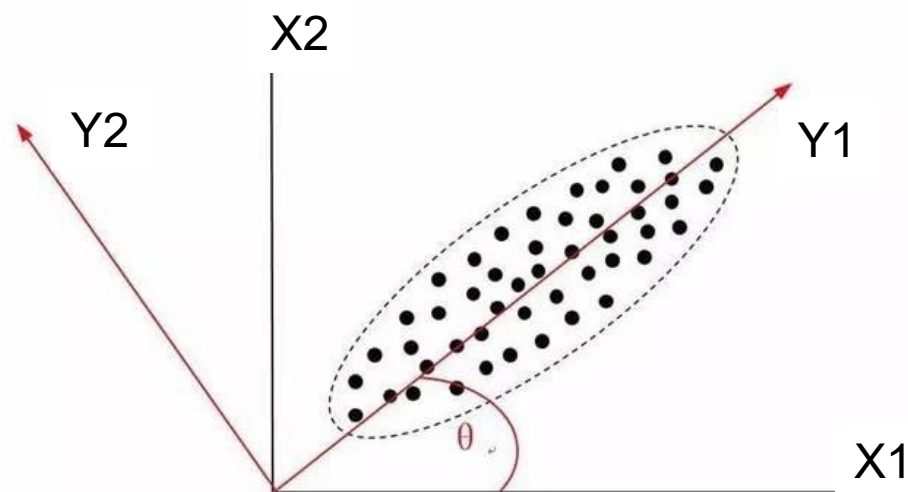
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$$y_i = \mathbf{u}_i^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad i = 1, 2$$



✓ 如何确定 u_{ij} : 方差最大化!

主成分分析

□ 数学推导

- 针对第一个主成分，令其方差最大化

$$\max_{\mathbf{u}_1} \frac{1}{n} \sum_{j=1}^n \left(\mathbf{u}_1^T \begin{bmatrix} x_{1j} \\ x_{2j} \end{bmatrix} - \mathbf{u}_1^T \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix} \right)^2$$



主成分分析

□ 数学推导

- 针对第一个主成分，令其方差最大化

目标函数

$$\max_{\mathbf{u}_1} \frac{1}{n} \sum_{j=1}^n \left(\mathbf{u}_1^T \begin{bmatrix} x_{1j} \\ x_{2j} \end{bmatrix} - \mathbf{u}_1^T \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix} \right)^2$$

约束条件

$$\mathbf{u}_1^T \mathbf{u}_1 = 1$$

- ✓ 等式约束优化问题

- 拉格朗日乘子法 (Lagrange multipliers)



主成分分析

□ 拉格朗日乘子法

- ① 假设目标函数是 $f(x, y)$, 限制条件是 $g(x, y) = 0$
- ② 构造拉格朗日函数 $F(x, y, \lambda) = f(x, y) + \lambda g(x, y)$
- ③ 对拉格朗日函数求偏导取0, 即 $\frac{\partial F}{\partial x} = \frac{\partial F}{\partial y} = \frac{\partial F}{\partial \lambda} = 0$



主成分分析

□ 数学推导

$$\max_{\mathbf{u}_1} \frac{1}{n} \sum_{j=1}^n \left(\mathbf{u}_1^T \begin{bmatrix} x_{1j} \\ x_{2j} \end{bmatrix} - \mathbf{u}_1^T \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix} \right)^2$$

$$\frac{1}{n} \sum_{j=1}^n \left(\mathbf{u}_1^T \begin{bmatrix} x_{1j} \\ x_{2j} \end{bmatrix} - \mathbf{u}_1^T \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix} \right)^2$$

$$= \frac{1}{n} \sum_{j=1}^n \left(\mathbf{u}_1^T \begin{bmatrix} x_{1j} \\ x_{2j} \end{bmatrix} - \mathbf{u}_1^T \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix} \right) \left(\mathbf{u}_1^T \begin{bmatrix} x_{1j} \\ x_{2j} \end{bmatrix} - \mathbf{u}_1^T \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix} \right)^T$$

$$= \mathbf{u}_1^T \frac{1}{n} \sum_{j=1}^n \left(\begin{bmatrix} x_{1j} \\ x_{2j} \end{bmatrix} - \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix} \right) \left(\begin{bmatrix} x_{1j} \\ x_{2j} \end{bmatrix} - \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix} \right)^T \mathbf{u}_1$$

$$= \mathbf{u}_1^T \begin{bmatrix} \frac{1}{n} \sum_{j=1}^n (x_{1j} - \bar{x}_1)^2 & \frac{1}{n} \sum_{j=1}^n (x_{1j} - \bar{x}_1)(x_{2j} - \bar{x}_2) \\ \frac{1}{n} \sum_{j=1}^n (x_{2j} - \bar{x}_2)(x_{1j} - \bar{x}_1) & \frac{1}{n} \sum_{j=1}^n (x_{2j} - \bar{x}_2)^2 \end{bmatrix} \mathbf{u}_1$$

主成分分析

□ 数学推导

$$\max_{\mathbf{u}_1} \frac{1}{n} \sum_{j=1}^n \left(\mathbf{u}_1^T \begin{bmatrix} x_{1j} \\ x_{2j} \end{bmatrix} - \mathbf{u}_1^T \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix} \right)^2$$

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$$\Sigma = \begin{bmatrix} \text{Var}(x_1) & \text{Cov}(x_1, x_2) \\ \text{Cov}(x_1, x_2) & \text{Var}(x_2) \end{bmatrix}$$

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主成分分析

□ 数学推导

➤ 带入拉格朗日函数

$$\Sigma = \begin{bmatrix} \text{Var}(x_1) & \text{Cov}(x_1, x_2) \\ \text{Cov}(x_1, x_2) & \text{Var}(x_2) \end{bmatrix}$$

$$F(\mathbf{u}_1, \lambda) = \mathbf{u}_1^T \Sigma \mathbf{u}_1 + \lambda(1 - \mathbf{u}_1^T \mathbf{u}_1)$$



主成分分析

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- 令偏导等于0

$$\frac{\partial F(\mathbf{u}_1, \lambda)}{\partial \mathbf{u}_1} = \Sigma \mathbf{u}_1 - \lambda \mathbf{u}_1 = 0$$

$$\Sigma \mathbf{u}_1 = \lambda \mathbf{u}_1$$



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特征值、特征向量!

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$$\Sigma \mathbf{u}_1 = \lambda \mathbf{u}_1$$

最大方差=最大特征值

$$\mathbf{u}_1^T \Sigma \mathbf{u}_1 = \mathbf{u}_1^T \lambda \mathbf{u}_1 = \lambda$$



主成分分析

□ 数学推导

- 协方差矩阵 Σ 是**实对称半正定阵**
 - ✓ 特征值 λ 是实数，特征向量 u 是实向量
 - ✓ 不同特征值的特征向量正交
- 协方差矩阵有多少阶，特征值和特征向量就有多少个
 - ✓ **第一主成分： λ 最大的特征向量**
 - ✓ **第二主成分： λ 第二大的特征向量**
 - ✓ 依此类推



主成分分析

□ 一般步骤

- 构建数据矩阵 $\mathbf{X}_{n \times p}$
- **对矩阵进行标准化处理**
- 建立协方差矩阵
- 求解协方差矩阵的特征值和特征向量
- 将特征值（即各主成分方差）从大到小排序

✓ 第 k 个主成分的方差贡献为 $\frac{\lambda_k}{\sum_{i=1}^p \lambda_i}$



主成分分析

□ 案例分析

A survey of 281 commuters was conducted in the Seattle metropolitan area. The intent of the survey was to gather information on commuters' opinions or attitudes regarding high-occupancy vehicle (HOV) lanes (lanes that are restricted for use by vehicles with two or more occupants). Commuters were asked a series of attitudinal questions regarding HOV lanes and their use, in addition to a number of sociodemographic and commuting behavior questions. A description of the variables is provided in [Table 8.1](#).

It is useful to assess whether a large number of variables, including attitudinal, sociodemographic, and travel behavior, can be distilled into a smaller set of variables. To do this, a principal components analysis is performed on the correlation matrix (not the variance-covariance matrix).



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主成分分析

□ 案例分析

Variable Abbreviation	Variable Description
CHGRTEPST5	On your past five commutes to work, how often have you changed route or departure time
HOVSAVTIME	HOV lanes save all commuters time: 0 if strongly disagree, 1 if disagree, 2 if neutral, 3 if agree, 4 if agree strongly
HOVADUSE	Existing HOV lanes are being adequately used: 0 if strongly disagree, 1 if disagree, 2 if neutral, 3 if agree, 4 if agree strongly
HOVOPN	HOV lanes should be open to all traffic: 0 if strongly disagree, 1 if disagree, 2 if neutral, 3 if agree, 4 if agree strongly
GPTOHOV	Converting some regular lanes to HOV lanes is a good idea: 0 if strongly disagree, 1 if disagree, 2 if neutral, 3 if agree, 4 if agree strongly
GTTOHOV2	Converting some regular lanes to HOV lanes is a good idea only if it is done before traffic congestion becomes serious: 0 if strongly disagree, 1 if disagree, 2 if neutral, 3 if agree, 4 if agree strongly
GEND	Gender: 1 if male, 0 if female
AGE	Age in years: 0 if under 21, 1 if 22 to 30, 2 if 31 to 40, 3 if 41 to 50, 4 if 51 to 64, 5 if 65 or older
HHINCM	Annual household income (U.S. dollars): 0 if no income, 1 if 1 to 9,999, 2 if 10,000 to 19,999, 3 if 20,000 to 29,999, 4 if 30,000 to 39,999, 5 if 40,000 to 49,999, 6 if 50,000 to 74,999, 7 if 75,000 to 100,000, 8 if over 100,000
EDUC	Highest level of education: 0 if did not finish high school, 1 if high school, 2 if community college or trade school, 3 if college/university, 4 if post college graduate degree
FAMSIZ	Number of household members
NUMADLT	Number of adults in household (aged 16 or older)
NUMWRKS	Number of household members working outside the home
NUMCARS	Number of licensed motor vehicles in the household
ZIPWRK	Postal zip code of workplace
ZIPHM	Postal zip code of home
HOVCMNT	Type of survey comment left by respondent regarding opinions on HOV lanes: 0 if no comment on HOV lanes, 1 if comment not in favor of HOV lanes, 2 if comment positive toward HOV lanes but critical of HOV lane policies, 3 if comment positive toward HOV lanes, 4 if neutral HOV lane comment

Variable Abbreviation	Variable Description
MODE	Usual mode of travel: 0 if drive alone, 1 if two person carpool, 2 if three or more person carpool, 3 if van pool, 4 if bus, 5 if bicycle or walk, 6 if motorcycle, 7 if other
HOVUSE	Have used HOV lanes: 1 if yes, 0 if no
HOVMODE	If used HOV lanes, what mode is most often used: 0 in a bus, 1 in two person carpool, 2 in three or more person carpool, 3 in van pool, 4 alone in vehicle, 5 on motorcycle
HOVDECLINE	Sometimes eligible for HOV lane use but do not use: 1 if yes, 0 if no
HOVDECREAS	Reason for not using HOV lanes when eligible: 0 if slower than regular lanes, 1 if too much trouble to change lanes, 2 if HOV lanes are not safe, 3 if traffic moves fast enough, 4 if forget to use HOV lanes, 5 if other
MODE1YR	Usual mode of travel 1 year ago: 0 if drive alone, 1 if two person carpool, 2 if three or more person carpool, 3 if van pool, 4 if bus, 5 if bicycle or walk, 6 if motorcycle, 7 if other
COM1YR	Commuted to work in Seattle a year ago: 1 if yes, 0 if no
FLEXSTAR	Have flexible work start times: 1 if yes, 0 if no
CHNGDEPTM	Changed departure times to work in the last year: 1 if yes, 0 if no
MINERLYWRK	On average, number of minutes leaving earlier for work relative to last year
MINLTWRK	On average, number of minutes leaving later for work relative to last year
DEPCHNGREAS	If changed departure times to work in the last year, reason: 0 if change in travel mode, 1 if increasing traffic congestion, 2 if change in work start time, 3 if presence of HOV lanes, 4 if change in residence, 5 if change in lifestyle, 6 if other
CHNGRTE	Changed route to work in the last year: 1 if yes, 0 if no
CHNGRTEREAS	If changed route to work in the last year, reason: 0 if change in travel mode, 1 if increasing traffic congestion, 2 if change in work start time, 3 if presence of HOV lanes, 4 if change in residence, 5 if change in lifestyle, 6 if other
I90CM	Usually commute to or from work on Interstate 90: 1 if yes, 0 if no
I90CMT1YR	Usually commuted to or from work on Interstate 90 last year: 1 if yes, 0 if no
HOVPST5	On your past five commutes to work, how often have you used HOV lanes
DAPST5	On your past five commutes to work, how often did you drive alone
CRPPST5	On your past five commutes to work, how often did you carpool with one other person
CRPPST52MR	On your past five commutes to work, how often did you carpool with two or more people
VNPPST5	On your past five commutes to work, how often did you take a van pool
BUSPST5	On your past five commutes to work, how often did you take a bus
NONMOTPST5	On your past five commutes to work, how often did you bicycle or walk
MOTPST5	On your past five commutes to work, how often did you take a motorcycle
OTHPST5	On your past five commutes to work, how often did you take a mode other than those listed in variables 18 through 24

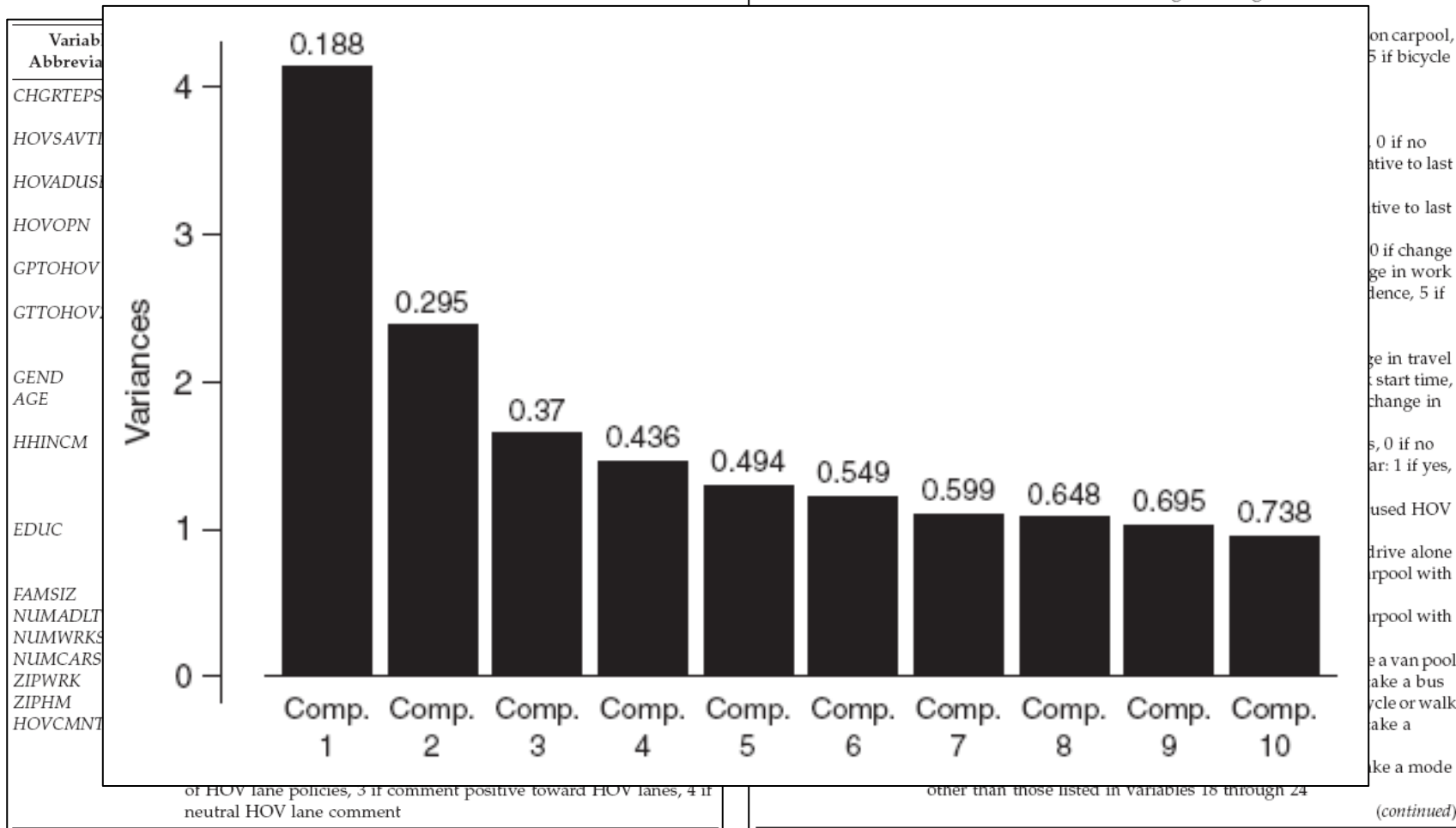
(continued)



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Factor Loadings of Principal Components Analysis: HOV Lane Survey Data

	Variable	Comp. 1	Comp. 2	Comp. 3	Comp. 4	Comp. 5	Comp. 6
	<i>Travel Behavior Variables</i>						
	HOVPST5	-0.380		-0.284	0.236		
V	DAPST5	0.396		0.274	-0.283		0.128
Abb	CRPPST5	-0.303		-0.223	0.240	0.282	0.221
CHGR	CRPPST52MR	-0.109			0.167	0.196	-0.107
HOVS	VNPPST5				-0.146		
HOVA	BUSPST5	-0.161	-0.140	-0.227	0.112	-0.514	-0.395
HOVC	NONMOTPST5						0.471
	MOTPST5			0.104		0.381	-0.418
GPTO	CHGRTEPST5					0.525	-0.302
	<i>HOV Attitude Variables</i>						
GTTO	HOVSAVTIME	-0.325		0.301	-0.140		
GEND	HOVADUSE	-0.321		0.227	-0.133		
AGE	HOVOPN	0.364		-0.216	0.210		
HHIN	GPTOHOV	-0.339	0.125	0.230	-0.115		
	GTTOHOV2	-0.260		0.245	-0.153		
	<i>Sociodemographic Variables</i>						
EDUC	GEND	0.117		0.388	0.180		-0.199
FAMS	AGE			0.268	0.341	-0.363	-0.270
NUM	HHINCM		0.304	0.131	0.489		0.101
NUM	EDUC		0.188	0.247	0.443		0.247
ZIPW	FAMSIZ		0.429	-0.122			
ZIPH	NUMADLT		0.516	-0.188	-0.128	-0.133	
HOVC	NUMWRKS		0.451	-0.242	-0.137		
	NUMCARS		0.372	-0.106		0.107	-0.268

Note: Loadings < 0.10 shown as blanks.



主成分分析与因子分析

□ 学习目标

- 主成分分析

- 因子分析



因子分析

□ 概念简介

- 用于分析**不能直接观测**的变量
 - ✓ 例如智力，爱国主义，消费者态度等
- 隐变量（因子）：不能直接观测，但可从**与可观测变量的相关关系**中推导出
 - ✓ 例如虚弱，可从力量、速度等可观测变量中推导出
 - ✓ **因子取值的变化，可影响多个可观测变量**



因子分析

□ 概念简介

➤ 探索性因子分析 (exploratory factor analysis)

- ✓ 降维：用少数几个因子来描述原变量的本质结构
- ✓ 无特定模型

➤ 验证性因子分析 (confirmatory factor analysis)

- ✓ 验证：某个因子模型对新样本的拟合程度
- ✓ 有特定模型



因子分析

□ 数学模型

- 给定 p 维观测变量 $\mathbf{X}$, 均值和协方差分别为 $\boldsymbol{\mu}$ 和 $\boldsymbol{\Sigma}$
- 假定 $\mathbf{X}$ 可表示为 m 个**公共因子** $F_1, \dots, F_m$ 和 p 个**特殊因子** $\epsilon_1, \dots, \epsilon_p$ 的**线性组合**

$$X_1 - \mu_1 = l_{11}F_1 + l_{12}F_2 + \cdots + l_{1m}F_m + \epsilon_1$$

$$X_2 - \mu_2 = l_{21}F_1 + l_{22}F_2 + \cdots + l_{2m}F_m + \epsilon_2$$

$$\vdots$$

$$X_p - \mu_p = l_{p1}F_1 + l_{p2}F_2 + \cdots + l_{pm}F_m + \epsilon_p$$

✓ l_{ij} : 第 i 个变量在第 j 个因子上的**因子负荷**



因子分析

□ 数学模型

- 给定 p 维观测变量 $\mathbf{X}$, 均值和协方差分别为 $\boldsymbol{\mu}$ 和 $\boldsymbol{\Sigma}$
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$$\vdots$$

$$X_p - \mu_p = l_{p1}F_1 + l_{p2}F_2 + \cdots + l_{pm}F_m + \epsilon_p$$

$$\mathbf{X}_{p \times 1} = \boldsymbol{\mu} + \mathbf{L}_{p \times m} \mathbf{F} + \boldsymbol{\epsilon}, \mathbf{L} \text{ 称为因子负荷阵}$$

✓ l_{ij} : 第 i 个变量在第 j 个因子上的**因子负荷**



因子分析

□ 数学模型

➤ 在**正交因子模型**的假设下

$$E\mathbf{F} = 0, \quad \text{Var}(\mathbf{F}) = E(\mathbf{F}\mathbf{F}') = I_m$$

$$E\boldsymbol{\epsilon} = 0, \quad \text{Var}(\boldsymbol{\epsilon}) = E(\boldsymbol{\epsilon}\boldsymbol{\epsilon}') = \Psi = \text{diag}\{\psi_1, \dots, \psi_p\}$$

$$\text{cov}(\mathbf{F}, \boldsymbol{\epsilon}) = 0$$



因子分析

□ 数学模型

➤ 在**正交因子模型**的假设下

$$E\mathbf{F} = 0, \quad \text{Var}(\mathbf{F}) = E(\mathbf{F}\mathbf{F}') = I_m$$

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$$\text{cov}(\mathbf{F}, \boldsymbol{\epsilon}) = 0$$

$$\text{Cov}(X_i, F_j) = l_{ij}$$

$$\Sigma = \text{Var}(\mathbf{X}) = \text{Var}(\mathbf{L}\mathbf{F} + \boldsymbol{\epsilon}) = \mathbf{L}\mathbf{L}' + \Psi$$

$$\text{Var}(\mathbf{L}\mathbf{F}) = E[\mathbf{L}\mathbf{F}(\mathbf{L}\mathbf{F})'] = E[\mathbf{L}\mathbf{F}\mathbf{F}'\mathbf{L}'] = \mathbf{L}E[\mathbf{F}\mathbf{F}']\mathbf{L}' = \mathbf{L}\mathbf{L}'$$



因子分析

$$\Sigma = \text{Var}(\mathbf{X}) = \text{Var}(\mathbf{LF} + \epsilon) = \mathbf{LL}' + \Psi$$

□ 数学模型

$$\sigma_{ii} = \text{Var}(X_i) = \sum_{j=1}^m l_{ij}^2 + \psi_i := h_i^2 + \psi_i$$

变量 $\mathbf{X}$ 的方差由两部分组成：

- ① m 个公共因子贡献的部分 h_i^2 称为**共性方差**
- ② 公共因子不能解释的部分，即由特殊因子解释的部分，称为**特殊方差**

因子分析力求**共性方差大，特殊方差小**！



因子分析

□ 数学模型

$$\Sigma = \text{Var}(\mathbf{X}) = \text{Var}(\mathbf{LF} + \epsilon) = \mathbf{LL}' + \Psi$$

$$\sigma_{ii} = \text{Var}(X_i) = \sum_{j=1}^m l_{ij}^2 + \psi_i := h_i^2 + \psi_i$$

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- ② 公共因子不能解释的部分，即由特殊因子解释的部分，称为**特殊方差**

$$S_j = \sum_{i=1}^p l_{ij}^2: \text{因子 } F_j \text{ 对变量 } \mathbf{X} \text{ 的总体方差贡献}$$



因子分析

□ 数学模型——案例

考虑如下协方差矩阵

$$\Sigma = \begin{bmatrix} 19 & 30 & 2 & 12 \\ 30 & 57 & 5 & 23 \\ 2 & 5 & 38 & 47 \\ 12 & 23 & 47 & 68 \end{bmatrix}$$

可得

$$\begin{aligned} \Sigma &= \begin{bmatrix} 4 & 1 \\ 7 & 2 \\ -1 & 6 \\ 1 & 8 \end{bmatrix} \begin{bmatrix} 4 & 7 & -1 & 1 \\ 1 & 2 & 6 & 8 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} \\ &= \mathbf{LL}' + \Psi. \end{aligned}$$



因子分析

□ 数学模型

➤ 公共因子 F 和负荷阵 L 不唯一!

任意正交矩阵

$$X - \mu = \mathbf{L}\mathbf{F} + \epsilon = (\mathbf{L}\mathbf{T})(\mathbf{T}'\mathbf{F}) + \epsilon = \mathbf{L}^*\mathbf{F}^* + \epsilon$$

- ✓ $\mathbf{L}^*$ 和 $\mathbf{F}^*$ 满足正交因子模型
- ✓ 负荷阵 $\mathbf{L}^* = \mathbf{L}\mathbf{T}$ 和 $\mathbf{L}$ 在解释协方差时等价
- ✓ 正交变换 → **因子旋转** → 旋转负荷阵, 得到易解释结果



因子分析

□ 负荷阵估计——主成分法

➤ 协方差矩阵 Σ 是**实对称半正定阵** → **正交对角化**

$$\Sigma = \mathbf{U} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_p \end{bmatrix} \mathbf{U}'$$

✓ $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$ 为 Σ 的特征值

✓ $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p$ 为对应的正交特征向量



因子分析

□ 负荷阵估计——主成分法

➤ 因此协方差矩阵 Σ 可做如下**精确分解**

$$\begin{aligned}
 \Sigma &= [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \cdots \quad \mathbf{u}_p] \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_p \end{pmatrix} \begin{bmatrix} \mathbf{u}'_1 \\ \mathbf{u}'_2 \\ \vdots \\ \mathbf{u}'_p \end{bmatrix} \\
 &= \lambda_1 \mathbf{u}_1 \mathbf{u}'_1 + \lambda_2 \mathbf{u}_2 \mathbf{u}'_2 + \cdots + \lambda_m \mathbf{u}_m \mathbf{u}'_m + \lambda_{m+1} \mathbf{u}_{m+1} \mathbf{u}'_{m+1} + \cdots + \lambda_p \mathbf{u}_p \mathbf{u}'_p \\
 &= [\sqrt{\lambda_1} \mathbf{u}_1 \quad \sqrt{\lambda_2} \mathbf{u}_2 \quad \cdots \quad \sqrt{\lambda_p} \mathbf{u}_p] \begin{bmatrix} \sqrt{\lambda_1} \mathbf{u}'_1 \\ \sqrt{\lambda_2} \mathbf{u}'_2 \\ \vdots \\ \sqrt{\lambda_p} \mathbf{u}'_p \end{bmatrix}
 \end{aligned}$$

因子分析

❑ 负荷阵估计——主成分法

- 寻求**少数几个公共因子** → 精确分解毫无价值
- 因此省略后面 $p - m$ 项

$$\begin{aligned}\Sigma &\approx \lambda_1 \mathbf{u}_1 \mathbf{u}_1' + \lambda_2 \mathbf{u}_2 \mathbf{u}_2' + \cdots + \lambda_m \mathbf{u}_m \mathbf{u}_m' + \hat{\mathbf{D}} \\ &= \begin{bmatrix} \sqrt{\lambda_1} \mathbf{u}_1 & \sqrt{\lambda_2} \mathbf{u}_2 & \cdots & \sqrt{\lambda_m} \mathbf{u}_m \end{bmatrix}_{p \times m} \begin{bmatrix} \sqrt{\lambda_1} \mathbf{u}_1' \\ \sqrt{\lambda_2} \mathbf{u}_2' \\ \vdots \\ \sqrt{\lambda_p} \mathbf{u}_m' \end{bmatrix}_{m \times p} + \hat{\mathbf{D}}\end{aligned}$$

其中 $\hat{\mathbf{D}} = \text{diag}(\hat{\sigma}_1^2, \hat{\sigma}_2^2, \dots, \hat{\sigma}_p^2)$

因子分析

□ 负荷阵估计——主成分法

➤ 正交因子模型的主成分分解

$$\Sigma \approx \mathbf{L}\mathbf{L}' + \text{diag}\{\psi_1, \dots, \psi_p\}$$

$$\mathbf{L} = [\sqrt{\lambda_1}\mathbf{u}_1, \dots, \sqrt{\lambda_m}\mathbf{u}_m]$$



因子分析

❑ 负荷阵估计——主成分法

➤ 正交因子模型的主成分分解

$$\Sigma \approx \mathbf{L}\mathbf{L}' + \text{diag}\{\psi_1, \dots, \psi_p\}$$



$$\mathbf{L} = [\sqrt{\lambda_1}\mathbf{u}_1, \dots, \sqrt{\lambda_m}\mathbf{u}_m]$$

$$\Psi = \text{diag}(\psi_1, \dots, \psi_p) = \text{diag}(\Sigma - \mathbf{L}\mathbf{L}')$$

$$\psi_i = s_{ii} - \sum_{j=1}^m l_{ij}^2$$



因子分析

□ 主成分法——案例

考虑如下**相关系数**矩阵

$$R = \begin{bmatrix} 1 & \frac{-1}{3} & 2/3 \\ \frac{-1}{3} & 1 & 0 \\ \frac{2}{3} & 0 & 1 \end{bmatrix}$$

➤ 首先求特征值

$$R - \lambda E = \begin{bmatrix} 1 - \lambda & \frac{-1}{3} & 2/3 \\ \frac{-1}{3} & 1 - \lambda & 0 \\ \frac{2}{3} & 0 & 1 - \lambda \end{bmatrix}$$

$$|R - \lambda X| = (1 - \lambda)^3 + 0 + 0 - (1 - \lambda)\frac{4}{9} - \frac{1}{9}(1 - \lambda) - 0 = 0$$



因子分析

□ 主成分法——案例

$$\lambda_1 = 1; \quad \lambda_2 = 1 - \frac{\sqrt{5}}{3} = 0.2546; \quad \lambda_3 = 1 + \frac{\sqrt{5}}{3} = 1.7454;$$

➤ 然后排序特征值，求特征向量

$$\lambda_1 = 1.7454; \quad \lambda_2 = 1; \quad v_1 = \begin{bmatrix} 0.7071 \\ 0.3162 \\ -0.6325 \end{bmatrix}; \quad v_2 = \begin{bmatrix} 0 \\ 0.8165 \\ 0.4082 \end{bmatrix}$$
$$A = [\sqrt{\lambda_1}v_1, \sqrt{\lambda_2}v_2] = \begin{bmatrix} 0.9342 & 0 \\ 0.4178 & 0.8165 \\ -0.8355 & 0.4082 \end{bmatrix}$$



因子分析

□ 主成分法——案例

$$\lambda_1 = 1; \quad \lambda_2 = 1 - \frac{\sqrt{5}}{3} = 0.2546; \quad \lambda_3 = 1 + \frac{\sqrt{5}}{3} = 1.7454;$$

➤ 然后排序特征值，求特征向量

✓ $\Sigma - LL'$ 非对角元较小

✓ $\frac{\sum_{i=1}^m \lambda_i}{\sum_{i=1}^p \lambda_i}$ 较大

✓ 相关系数矩阵特征值不小于1

$$\lambda_1 = 1.7454; \quad \lambda_2 = 1; \quad v_1 = \begin{bmatrix} 0.7071 \\ 0.3162 \\ -0.6325 \end{bmatrix}; \quad v_2 = \begin{bmatrix} 0.8165 \\ 0.4082 \end{bmatrix}$$

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因子分析

□ 因子旋转

➤ 公共因子 F 和负荷阵 L 不唯一!

$$X - \mu = LF + \epsilon = (LT)(T'F) + \epsilon = L^*F^* + \epsilon$$



因子分析

□ 因子旋转

➤ 公共因子 F 和负荷阵 L 不唯一！

$$X - \mu = LF + \epsilon = (LT)(T'F) + \epsilon = L^*F^* + \epsilon$$

➤ 正交阵 $T \rightarrow$ 因子旋转 $\rightarrow$ **因子负荷平方两极分化**，即要么接近0，要么接近1

✖ 公共因子对许多变量有较大负荷，不利于解释

✓ 公共因子对部分变量有较大负荷，有利于解释



因子分析

□ 因子旋转

最大方差正交旋转法

$$\begin{aligned} X_1 - \mu_1 &= l_{11}F_1 + l_{12}F_2 + \cdots + l_{1m}F_m + \epsilon_1 \\ X_2 - \mu_2 &= l_{21}F_1 + l_{22}F_2 + \cdots + l_{2m}F_m + \epsilon_2 \\ &\vdots \\ X_p - \mu_p &= l_{p1}F_1 + l_{p2}F_2 + \cdots + l_{pm}F_m + \epsilon_p \end{aligned}$$

- 通过**因子旋转**，调整负荷阵的**列**，使变量在每个因子上的负荷尽可能**分散**，即（列）**方差最大**
- 选择 **T** 使如下 **V** 最大

$$V = \sum_{j=1}^m \left[\frac{1}{p} \sum_{i=1}^p \left\{ (l_{ij}^*)^2 - \frac{1}{p} \sum_{i=1}^p (l_{ij}^*)^2 \right\}^2 \right]$$

✓ 其中 $(l_{ij}^*)^2 = \frac{(\hat{l}_{ij}^*)^2}{(\hat{h}_i)^2} = \frac{(\hat{l}_{ij}^*)^2}{\sum_{j=1}^m (\hat{l}_{ij}^*)^2}$



因子分析

□ 因子旋转

最大方差正交旋转法

$$\begin{aligned} X_1 - \mu_1 &= l_{11}F_1 + l_{12}F_2 + \cdots + l_{1m}F_m + \epsilon_1 \\ X_2 - \mu_2 &= l_{21}F_1 + l_{22}F_2 + \cdots + l_{2m}F_m + \epsilon_2 \\ &\vdots \\ X_p - \mu_p &= l_{p1}F_1 + l_{p2}F_2 + \cdots + l_{pm}F_m + \epsilon_p \end{aligned}$$

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✓ **V** 是旋转角度 θ 的函数：单变量优化问题



因子分析

□ 因子得分

计算公共因子 F 的数值：**回归估计法**

- 假设公共因子对 p 个变量做回归

$$\hat{F}_j = b_{j0} + b_{j1}X_1 + \cdots + b_{jp}X_p \quad (j = 1, \cdots, m)$$

$$b_{j0} = 0$$



因子分析

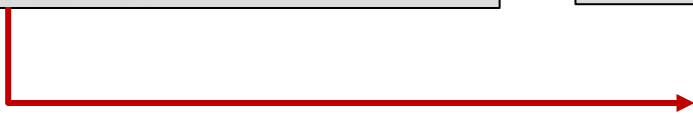
□ 因子得分

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$$b_{j0} = 0$$


$$\hat{F} = BX$$



因子分析

□ 因子得分

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- 假设公共因子对 p 个变量做回归

$$\hat{F}_j = b_{j0} + b_{j1}X_1 + \cdots + b_{jp}X_p \quad (j = 1, \cdots, m)$$

$$b_{j0} = 0$$

$$\hat{\mathbf{F}} = \mathbf{B}\mathbf{X}$$

- 考虑去中心化 X_i

$$\begin{aligned} l_{ij} &= \text{Cov}(X_i, F_j) = E(X_i F_j) \\ &= E[X_i(b_{j1}X_1 + \cdots + b_{jp}X_p)] \\ &= b_{j1}E(X_i X_1) + \cdots + b_{jp}E(X_i X_p) \\ &= b_{j1}\sigma_{i1} + \cdots + b_{jp}\sigma_{ip} \end{aligned}$$



因子分析

□ 因子得分

计算公共因子 F 的数值: **回归估计法**

- 假设公共因子对 p 个变量做回归

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$$\Sigma = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1p} \\ \vdots & \ddots & \vdots \\ \sigma_{p1} & \cdots & \sigma_{pp} \end{bmatrix}, \quad \Sigma \begin{bmatrix} b_{j1} \\ \vdots \\ b_{jp} \end{bmatrix} = \begin{bmatrix} l_{1j} \\ \vdots \\ l_{pj} \end{bmatrix}$$



因子分析

□ 因子得分

$$\Sigma = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1p} \\ \vdots & \ddots & \vdots \\ \sigma_{p1} & \cdots & \sigma_{pp} \end{bmatrix}, \quad \Sigma \begin{bmatrix} b_{j1} \\ \vdots \\ b_{jp} \end{bmatrix} = \begin{bmatrix} l_{1j} \\ \vdots \\ l_{pj} \end{bmatrix}$$


$$\Sigma \begin{bmatrix} b_{11} & \cdots & b_{m1} \\ \vdots & \ddots & \vdots \\ b_{1p} & \cdots & b_{mp} \end{bmatrix} = \begin{bmatrix} l_{11} & \cdots & l_{1m} \\ \vdots & \ddots & \vdots \\ l_{p1} & \cdots & l_{pm} \end{bmatrix}$$

因子分析

□ 因子得分

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$$\Sigma B' = L$$



因子分析

□ 因子得分

$$\Sigma = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1p} \\ \vdots & \ddots & \vdots \\ \sigma_{p1} & \cdots & \sigma_{pp} \end{bmatrix}, \quad \Sigma \begin{bmatrix} b_{j1} \\ \vdots \\ b_{jp} \end{bmatrix} = \begin{bmatrix} l_{1j} \\ \vdots \\ l_{pj} \end{bmatrix}$$

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$$\Sigma \mathbf{B}' = \mathbf{L}$$

$$\hat{\mathbf{F}} = \mathbf{B}\mathbf{X} = \mathbf{L}'\Sigma^{-1}\mathbf{X}$$

若数据标准化, 协方差矩阵即为相关系数矩阵



因子分析

□ 步骤总结

➤ 检验变量相关性

- ✓ KMO 检验、Bartlett 球形检验

➤ 估计负荷阵

- ✓ 因子负荷、因子旋转

➤ 模型解释

- ✓ 负荷分布、因子得分



参考文献

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Statistical AND Econometric Methods FOR Transportation Data Analysis

**Simon P. Washington
Matthew G. Karlaftis
Fred L. Mannering**



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